

Ex. No: 5. Use Autopsy to create a case and import evidence

Date:

Description

Autopsy is an open-source digital forensics platform used for analyzing and extracting data from digital devices. Here's a step-by step process on how to use Autopsy for a basic forensic investigation: Drive link:

https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8_yJ8kixc_xTU-v

Download files: 4Dell Latitude CPi.E01, 4Dell Latitude CPi.E02

1. Installation

Download and Install: Autopsy can be downloaded from the official website. Follow the installation instructions based on your operating system (Windows, Linux, or macOS).

2. Starting a New Case

Open Autopsy: Launch the application after installation. ·

Create a New Case:

- Enter the case name and location where the case data will be stored.
- Fill in the details like the case number, examiner's name, etc., and click Next.

New Case Information

Steps

1. Case Information
2. Optional Information

Case Information

Case Name: Yaswanth

Base Directory: C:\Users\DELL\OneDrive\ドキュメント Browse

Case Type: ☒ Single-User ☐ Multi-User

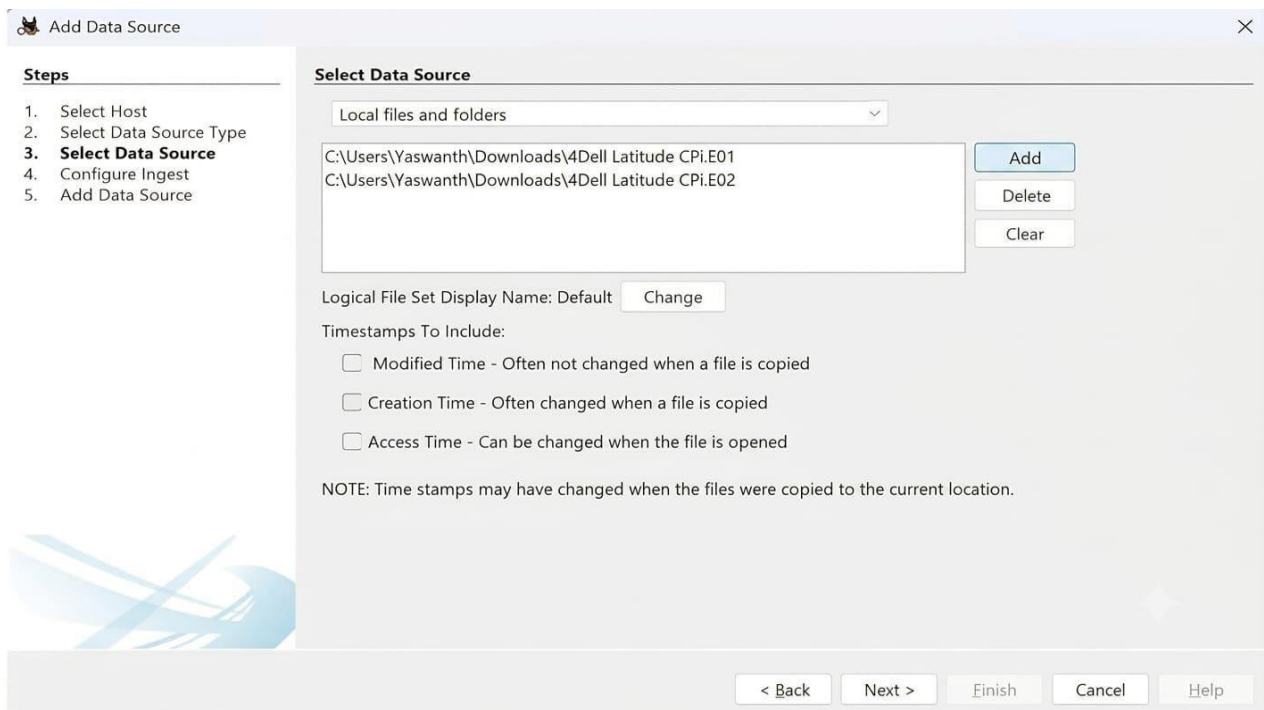
Case data will be stored in the following directory:

C:\Users\DELL\OneDrive\ドキュメント\Yaswanth

< Back Next > Finish Cancel Help

3. Adding a Data Source

- Choose the Type of Data Source:
- After creating a case, you'll be prompted to add a data source.
- You can add disk images, directories, logical files, or local disks.
Select the Data Source:
- Browse to the location of the image file (e.g., E01, dd, or raw), physical disk, or directory you want to analyze. https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v



Configure Ingest Modules:

- Autopsy allows you to select specific analysis modules such as File Type Identification, Keyword Search, Hash Lookup, etc.
- You can enable or disable these based on your investigation needs.
- Start Analysis: Click Next to start the analysis.

4. Initial Analysis and Overview

- Ingest Progress: As Autopsy processes the data source, you'll see the progress in the lower-left corner.

Explore the Resulting Artifacts:

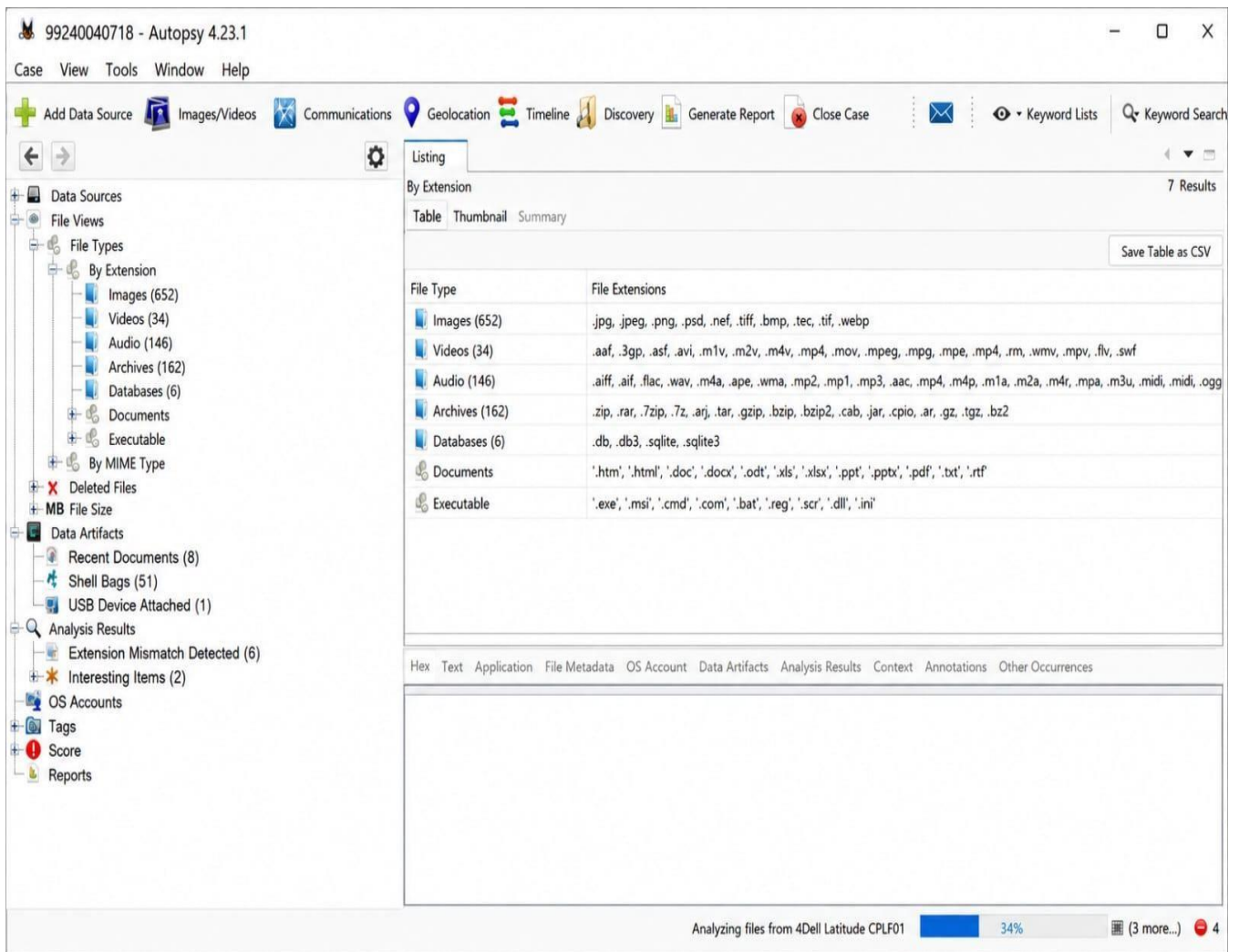
- Autopsy automatically categorizes findings such as web artifacts, file system metadata, and communication records.

Use the Tree Viewer:

- On the left pane, you'll see a tree structure where you can explore different aspects like File System, Web History, Email, etc.

5. Detailed Analysis

· Keyword Search:



· You can perform specific keyword searches using the Keyword Search module.

· Use pre-configured lists or enter custom keywords. ·

File Analysis:

· Navigate through files and folders under the File Types or File System section. ·

Open, view, or export files for further examination. ·

Timeline Analysis:

· Use the Timeline module to visualize events based on timestamps. ·

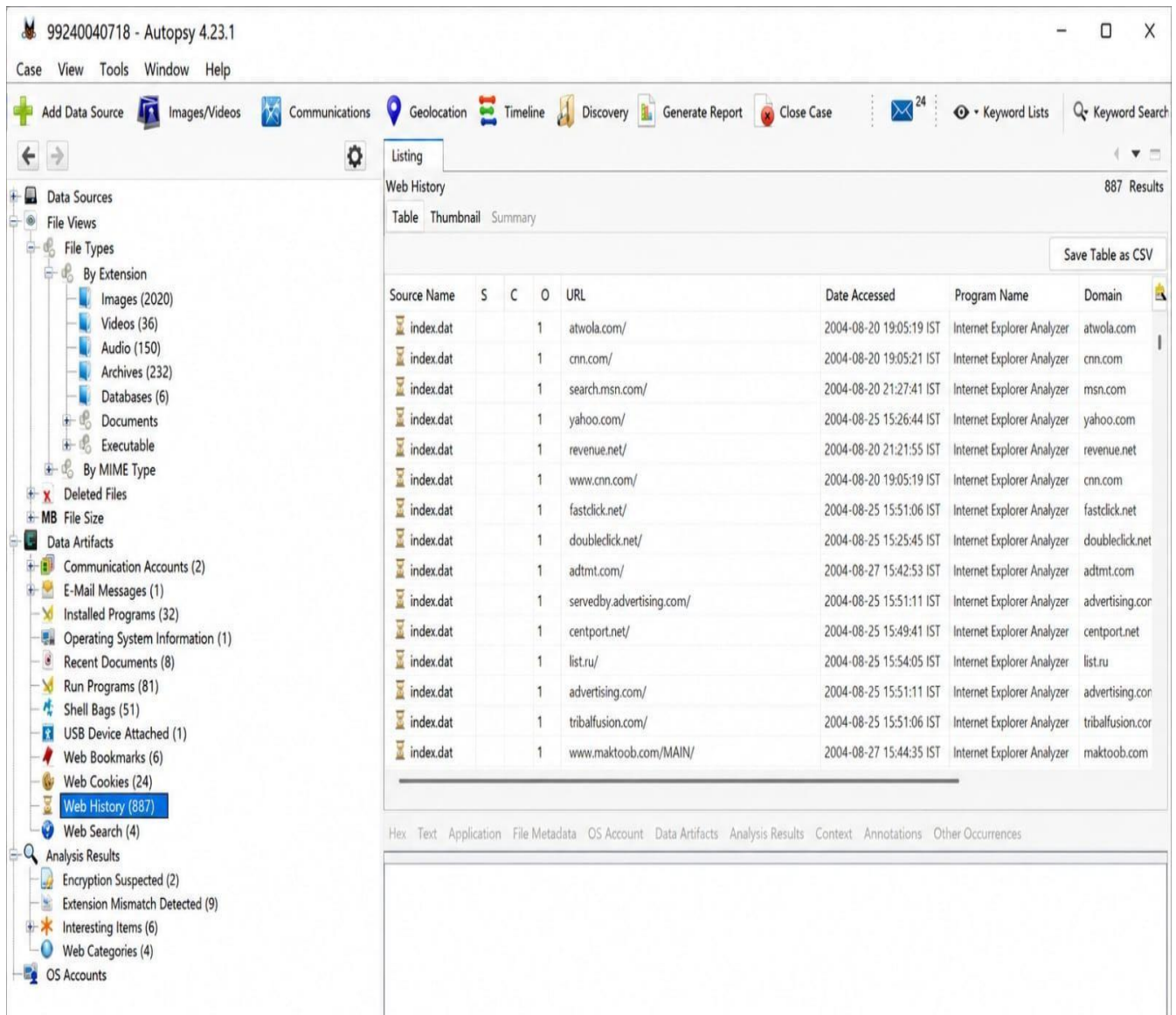
This can help track user activity over time.

· Hash Analysis:

· Compare file hashes with known databases to identify known good or bad files.

6. Reporting

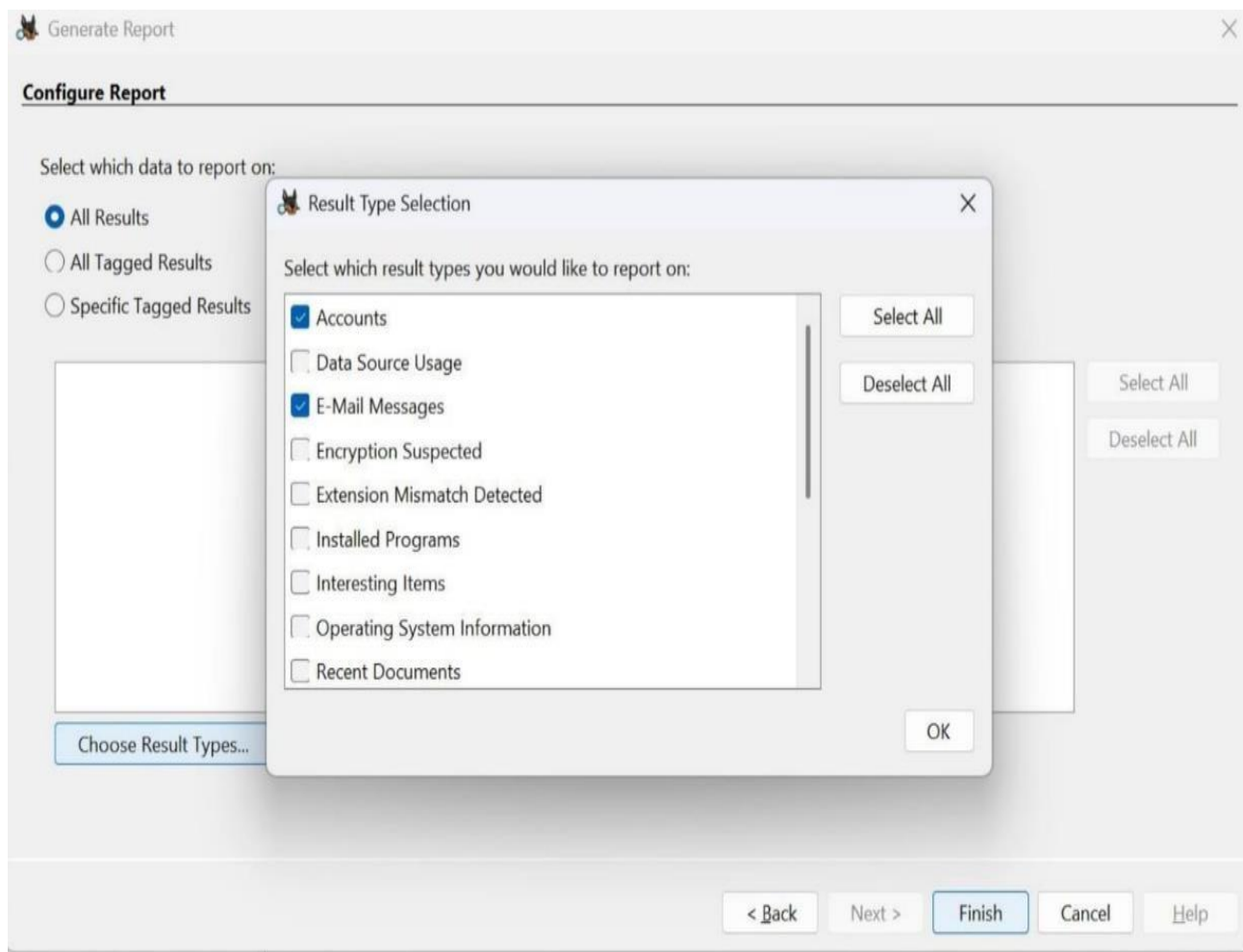
- Generate a Report:



The screenshot shows the Autopsy 4.23.1 interface. The left sidebar displays a tree view of data sources, with 'Web History (887)' selected. The main pane shows a 'Listing' of web history entries. The table has columns: Source Name, S, C, O, URL, Date Accessed, Program Name, and Domain. The data shows multiple entries for 'index.dat' files, all accessed by 'Internet Explorer Analyzer' from various domains like atwola.com, cnn.com, msn.com, yahoo.com, revenue.net, and www.cnn.com. The bottom of the interface shows tabs for 'Hex', 'Text', 'Application', 'File Metadata', 'OS Account', 'Data Artifacts', 'Analysis Results', 'Context', 'Annotations', and 'Other Occurrences'.

Source Name	S	C	O	URL	Date Accessed	Program Name	Domain
index.dat			1	atwola.com/	2004-08-20 19:05:19 IST	Internet Explorer Analyzer	atwola.com
index.dat			1	cnn.com/	2004-08-20 19:05:21 IST	Internet Explorer Analyzer	cnn.com
index.dat			1	search.msn.com/	2004-08-20 21:27:41 IST	Internet Explorer Analyzer	msn.com
index.dat			1	yahoo.com/	2004-08-25 15:26:44 IST	Internet Explorer Analyzer	yahoo.com
index.dat			1	revenue.net/	2004-08-20 21:21:55 IST	Internet Explorer Analyzer	revenue.net
index.dat			1	www.cnn.com/	2004-08-20 19:05:19 IST	Internet Explorer Analyzer	cnn.com
index.dat			1	fastclick.net/	2004-08-25 15:51:06 IST	Internet Explorer Analyzer	fastclick.net
index.dat			1	doubclick.net/	2004-08-25 15:25:45 IST	Internet Explorer Analyzer	doubclick.net
index.dat			1	admt.com/	2004-08-27 15:42:53 IST	Internet Explorer Analyzer	admt.com
index.dat			1	servedby.advertising.com/	2004-08-25 15:51:11 IST	Internet Explorer Analyzer	advertising.com
index.dat			1	centport.net/	2004-08-25 15:49:41 IST	Internet Explorer Analyzer	centport.net
index.dat			1	list.ru/	2004-08-25 15:54:05 IST	Internet Explorer Analyzer	list.ru
index.dat			1	advertising.com/	2004-08-25 15:51:11 IST	Internet Explorer Analyzer	advertising.com
index.dat			1	tribalfusion.com/	2004-08-25 15:51:06 IST	Internet Explorer Analyzer	tribalfusion.com
index.dat			1	www.maktoob.com/MAIN/	2004-08-27 15:44:35 IST	Internet Explorer Analyzer	maktoob.com

- After analyzing the data, click on Generate Report from the toolbar.
 - Choose the type of report (HTML, CSV, Excel, etc.).
- Select which parts of the analysis you want to include in the report.
 - Export Findings:
- Export individual files or artifacts that you need for your report or further analysis.
 - Final Review:
- Review the report to ensure it includes all relevant information.
 - Save or print the report for use in your case.



7. Case Closure

- Close the Case:
- Once you have completed your investigation, close the case within Autopsy.

Archiving:

- Ensure all data and reports are properly archived according to your organization's policies.

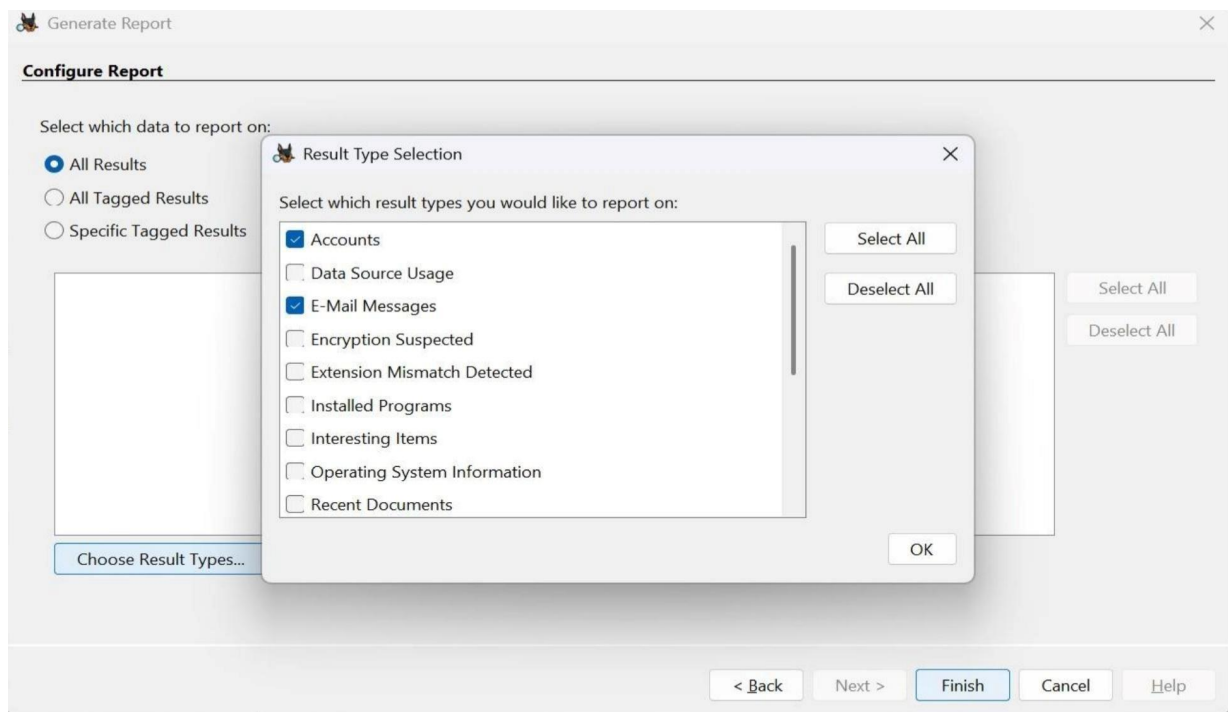
8. Advanced Features (Optional)

Custom Ingest Modules:

- Autopsy allows for custom modules to be added if you need specific analysis tools not included by default.

Collaboration:

- Autopsy can be configured for multi-user cases if you're working in a team environments



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT

- The Autopsy case was successfully created and the evidence files were imported. The forensic data was analyzed and the required findings were reviewed.
- The selected results were compiled and the forensic report was generated successfully.
- Thus, the investigation and reporting process was completed successfully.